

Engine Inspection Brief — Field Checklist

Purpose: close every open unknown that no amount of desk research can close — only a person with a multimeter and a camera, at the machine, can. Closes or advances U1, U2, U4, U6, U7, U8, U9, U10, U11, U12, plus the pin 01/02 ground question raised in the reconciled spec.

Print this. Carry it. Tick boxes as you go — a half-finished visit is still useful if the ticks say which half.

Sources: reconciled spec `docs/superpowers/specs/2026-09-13-ecu-reconciled-spec-design.md` (§3 safety, §8 unknowns register), `refs/ecu-pinout-extracted.md`, research memos `docs/research/01-engine-identity.md` and `docs/research/03-ecu-connector.md`. Every expected reading below cites one of these. Anything not sourced is marked **inferred**.

SAFETY — read this before touching anything

This is a 25 kVA genset: a diesel engine that can start, and a 415 V three-phase alternator. This checklist is bench/inspection work only.

- **Battery disconnected before any continuity or resistance measurement.** Negative terminal off first, isolated, out of reach of the positive post.
- **Nothing inside the alternator output enclosure.** No task here requires it. If you find yourself reaching toward the 415 V output terminals, stop — that is not this checklist.
- **The engine does not run during any of this.** Every task below is done with the engine off. Ignition-on tasks mean the key/master switch position only, never cranking or running.
- **If a measurement needs the set running** (this includes the full CAN log in Group 4, which may need it), **that is a separate supervised activity**, done by someone qualified to run the set, not a solo checklist item. Note it as such and move on.
- Reconnect the battery only after Group 1 is fully complete and you've checked no tools or leads are left on the terminals.

Materials: digital multimeter (V, Ω , continuity beep), clamp meter if available (safer than breaking a fused circuit for current), phone/camera with flashlight, notebook or voice memo, 10 mm socket for the battery terminal, tape/marker for labelling. **Group 4 additionally needs a USB-CAN adapter and a laptop** — it may warrant its own trip.

Group 1 — Battery disconnected (continuity / resistance)

Do these first, in this order, with the battery negative terminal removed and isolated.

- ▣ **1.1 — Pin 01/02 ground hypothesis.** Continuity from ECU connector pins 01 and 02 to battery negative / chassis ground.
 - *Why:* the connector drawing places 01/02/05/06 as large power contacts, but 01/02 are labelled `G_G_B AT1 / AT2` — switch-return legs, which wouldn't need a large contact. The alternate reading: `G_G_B` = battery ground, and these are the ECU's main power grounds (also the only explanation for why no power-ground pin appears anywhere in the 44-pin extraction). (*spec §4 note; pinout "Digital inputs"*)
 - **Continuity (near 0 Ω) → hypothesis confirmed:** 01/02 are power grounds, not switch returns. Update the pin map and power/ground plan.
 - **Open circuit → hypothesis wrong.** They're isolated switch-return legs as originally labelled; the power-ground pin(s) remain unidentified — keep looking during Group 3 tracing.
- [] **1.2 — Pin 09 "synchronization ground."** Continuity from pin 09 to (a) battery negative and (b) a known sensor ground pin (e.g. 34 or 08).
 - *Why:* pin 09 is named as a ground but drawn in **red** (switched-battery colour class), which is contradictory on its face. (*spec §4, "Power"; pinout, U6 entry*)
 - **Continuous to battery negative** → it likely is a ground point, oddly coloured in the diagram for harness-routing reasons, not a live signal. Note this for §4 but treat as resolved.
 - **Open to both** → it is *not* a simple ground. Carry to Group 2 (2.3) for a voltage measurement — the red wire colour then means what it usually means.
- [] **1.3 — Boost sensor thermistor topology (closes U2 outright).** Boost P&T sensor unplugged, engine **cold**. Resistance across sensor pins 2 and 1 (ECU pins 79 / 34 per the pinout — temperature signal / sensor ground). (*brief; spec §8 U2; pinout "Analog sensors"*)
 - **A few k Ω , falling if you re-check once the engine's warmed a little** → NTC thermistor. Design a pull-up divider front-end. **U2 closed.**
 - **Open circuit, or a fixed high reading that doesn't move with temperature** → not a bare NTC — a conditioned/ratiometric output. Design the ratiometric front-end (same as the pressure channel). **U2 closed.**
 - This is a single measurement. Either outcome closes U2 — there is no third case.

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- ▣ **1.4 — Ignition relay contact type. Do this BEFORE task 2.1; it decides how 2.1's result is read.** Locate ignition relay `-13RB1` on the GCU relay board. With the battery disconnected and the relay coil unpowered, measure continuity across its **load contacts** (not the coil).

Why this matters and why 2.1 is ambiguous without it: task 2.1 reads voltage with the ignition switch off, assuming that is the relay's de-energised state. If the relay turns out to be normally-closed, "de-energised" means contacts **made** and power still present — so a battery-voltage reading in 2.1 would mean the test state was wrong, not that the relay only signals. Those two have opposite conclusions and the same reading. This measurement separates them. **Inferred concern, not sourced — raised because the relay's drive logic is not documented anywhere we have.**

Continuity across load contacts, coil unpowered	Meaning for task 2.1
Open (normally-open, energise to run)	Expected case. 2.1's table reads as written.
Closed (normally-closed, energise to cut)	2.1's table is inverted . Battery voltage with the switch off is then normal, and U12 must instead be tested by energising the relay and re-measuring. Note it and flag it.

Group 2 — Ignition switch tests (battery connected, engine OFF, not cranking)

Reconnect the battery for this group only. Key/master switch is the only thing that moves. **Do not crank or start.**

- ▣ **2.1 — U12, the priority item. Complete task 1.4 first.** Ignition/master switch **OFF**. This is assumed to be ignition relay `-13RB1`'s de-energised state — **task 1.4 confirms that assumption, and if it fails, the table below inverts.** You do not need to pull or force the relay. If you can see or hear the relay (GCU relay board), confirm it looks/sounds unenergised before measuring; photograph its label to confirm 70 A rating.

With the switch OFF, measure voltage (relative to chassis/battery negative) at: - ECU connector pin **21** (battery+ feed) - ECU connector pin **04** (`V_V_BAT_1R`) - ECU connector pin **06** (`V_V_BAT_2R`) - the **fuel metering unit's own pin 2** (its battery-fed pin, via 20 A fuse `8F1` — not ECU pin 88, which is the ECU's PWM output to metering unit pin 1)

Why this is the whole point of the trip: spec §3's safety argument — that a hung ECU can't keep injecting because the GCU physically removes fuel-system power — depends entirely on this. (*spec §3, §8 U12*)

Reading at all four points	Meaning
0 V (no battery voltage present)	Relay removes power . §3's second line of defence is real. U12 closed in the safe direction.
Battery voltage present at any of them	Relay signals only — power stays up through a hung ECU. §3's relaxation is void: reverts to "standalone MPU trip module mandatory." Flag this immediately, do not wait for the writeup.

- ▣ 2.2 — U7: are pins 04/06 sense-only or load feeds? Switch ON (engine still off). Re-measure voltage at pins 04 and 06 (expect battery voltage present now, per 2.1's contrast — that alone doesn't answer U7).
- **Then measure current**, preferably with a clamp meter on the wire (safer — no circuit break). If only a multimeter is available, break the circuit at the fuse holder for 04/06 and measure DC current in series, briefly.
- *Why*: sense lines draw milliamps; something that's actually feeding a load draws amps, and each fuse is rated 10 A. (*brief; spec §8 U7; spec §4 "Power" — both pins are 10 A fused*)

Current	Meaning
mA range	Sense-only — these just feed the battery-voltage ADC divider. Power sheet needs no extra current rating here.
Amps, approaching the 10 A fuse rating under some load state	Load feed — something downstream draws real current through the ECU. Power sheet must size for it.

- ▣ 2.3 — Pin 09, if 1.2 left it open. Switch ON. Voltage at pin 09 vs. chassis.
- ~0 V → confirms a ground reference despite the odd wire colour.
- **Switched battery voltage** → it's a real signal wearing a "ground" name — treat as an undocumented discrete/analog input, not a ground. Note whatever else in the harness moves when this changes (if observable) as a clue to its function.

Group 3 — Photography (engine off; battery may be either state, disconnect

preferred while working around the engine bay)

For each: square-on unless noted, fill the frame, flash off / raking light for part numbers (on-axis flash washes out shallow embossing).

- ▣ 3.1 **Engine rating plate** (rocker cover / flywheel housing / block side). Get legible: model string (expect something matching **3R550ETA** — anything else means the engine-identity inference in memo 01 is wrong and should be discarded, not patched), serial number, rated output/speed/frequency, any emissions certification mark. **Closes U11.** (*memo 01, "What would close this"*)

- ▣ **3.2 Genset nameplate** (canopy/control panel). Expect **KG4-25WS1** or close variant, plus genset serial number. (*memo 01*)
- ▣ **3.3 ECU case label**. Flat, well-lit. Hunting a vendor part number — Bosch EDC pattern **0 281 0xx xxx**, Bosch-reseller pattern **F 01R 0xx xxx**, or any Continental/Delphi/Denso mark. Would resolve the still-open OEM-ECU-vendor question. (*memo 03 §4 item 4*)
- ▣ **3.4 ECU connector — full-face, mating side, connector unplugged**. Determines at a glance: flat rectangular multi-row grid (Bosch/Delphi-style) vs. oval/rounded single-lever shell (AMPSEAL-style) — the top-level branch of the connector shortlist. **Advances U1**. (*memo 03 §4 item 1, ranked candidate table*)
- ▣ **3.5 Connector retention mechanism**, actuated and disengaged, if it can be operated safely with the connector unplugged. Lever vs. slide latch vs. bolt vs. friction/CPA clip. (*memo 03 §4 item 2*)
- ▣ **3.6 Raking-light pass over every flat face of the connector housing** — top, both sides, cable boot. Hunting molded/raised part number, date code, or logo: Bosch roundel, Delphi/Aptiv wordmark, TE Connectivity mark. Take several shots at different light angles if the first doesn't show text. **Would settle U1's connector-family ranking outright if a logo turns up**. (*memo 03 §4 item 3*)
- ▣ **3.7 Connector still mated, wide shot** — backshell, cable exit angle, strain relief. Feeds the adapter-harness design even without a family ID. (*memo 03 §4 item 5*)
- ▣ **3.8 Pin-count/contact-size check, connector unplugged**. Confirm four rows of small pins plus two 2-pin corner blocks (near pins 1 and 5). **Check specifically whether the corner-block contacts (pins 1/2/5/6) are visibly larger/wider than the small-pin grid**. This is the photographic half of the 1.1 pin 01/02 question — larger contacts support "these are power," normal-sized contacts support "these are ordinary switch returns after all." (*memo 03 §4 item 6; spec §4 note*)
- ▣ **3.9 Boost P&T sensor body** — part number, if legible. Backs up 1.3's electrical answer with a datasheet-checkable part.
- ▣ **3.10 Injector body** — part number, if accessible without disturbing the fuel system. Corroborates the Bosch-family injector inference already used in memo 03/04.
- ▣ **3.11 Intake throttle — 6-pin connector and cable run (closes U8)**. Photograph the connector, then follow the cable by hand through the loom.
- **Terminates at the ECU's 94-pin connector** → in scope. Photograph which physical position on the ECU housing it plugs into (row/corner), so it can be matched to a pin number once U1 is resolved.
- **Joins the XC11/XC1 panel loom instead** (same path as fuel level/WIF, per spec §2.3) → out of scope for this ECU; a throttle driver sheet is not needed.

- ▣ 3.12 Catalyst temperature sensor — 2-pin connector and harness run (closes U9). Same method as 3.11: photograph, then trace.
 - Terminates at ECU → in scope, an EGT-style front-end is needed.
 - Joins the panel/GCU loom → out of scope, same treatment as fuel level/WIF.
- ▣ 3.13 KG640C terminal strip (J1–J8), photographed with terminal labels legible. Reference material for the MPU-input question (U3) and future GCU interface work; not expected to close anything by itself this visit.
- ▣ 3.14 Wire-colour spot check at each analog sensor connector (boost, coolant, EGR, oil pressure, rail pressure) — flag toward U10. SENT and CAN H share the same yellow colour code in this harness's legend (*pinout*, "*Wire colour classes*"), so this can't be resolved by colour alone. Photograph any single yellow wire that is not paired with a green wire (a CAN pair) and is not part of a 3-wire AC bundle — that's the SENT candidate. This will likely only narrow it down, not close it outright — say so in the notes rather than guessing a pin.

Group 4 — Separate trip: CAN bus log (closes U4 in practice, not from a manual)

Needs a USB-CAN adapter and a laptop. Do this as its own visit unless you're already bringing the hardware.

- ▣ 4.1 With the OEM ECU still fitted and the harness intact, tap CAN channel 1 (pins 55/77, H/L) and channel 2 (pins 87/86, H/L) via breakout or fly leads. (*pinout*, "*CAN — two channels*"; *spec §4*)
- ▣ 4.2 Log traffic with the ignition on (ECU and GCU powered). Identify PGNs present, source addresses, and update rates.
 - Basic identification/status broadcasts should appear at key-on. **Engine-state PGNs (speed, load) may not appear, or may read zero/stale, until the engine is actually running** — if you need those, that's the "separate supervised activity" from the safety preamble, not a solo add-on to this trip.
- ▣ 4.3 Compare the captured PGN set against whatever the KG640C is found to expect (from the log itself, and cross-check against the KG640 manual if it surfaces). This is the practical closure the brief couldn't get from documents — **the log is the source, not a spec.** (*spec §8 U4*)

What this closes

#	Unknown	Closed by
U1	Connector family	3.4, 3.5, 3.6, 3.7, 3.8 (photographic — narrows the ranked shortlist in memo 03, may not fully settle it without a logo hit)

#	Unknown	Closed by
U2	Boost air-temp topology	1.3 — closes outright
U4	KG640C PGN set	Group 4 — practical answer via live log
U6	Pin 09 function	1.2, 2.3
U7	Pins 04/06 sense vs. load	2.2
U8	Intake throttle on ECU?	3.11
U9	Catalyst sensor routing	3.12
U10	Which signal is SENT	3.14 — narrows, may not fully close
U11	Engine identity	3.1 — closes outright if plate matches
U12	Ignition relay: power-cut or signal-only	2.1 — the safety-gate item; closes §3's open question either way
—	Pin 01/02 ground hypothesis	1.1, 3.8

Ten register entries touched, two (U2, U11) closeable outright on the spot, one (U12) resolves the live safety-argument branch either way, the rest narrowed with a clear next step recorded regardless of outcome.